

University of Bahrain

College of Information Technology
Department of Computer Science

ITCS347 Design and Analysis of Algorithms

Exercises #2 – *Fundamentals of the Analysis of Algorithm Efficiency*

1. Consider the following program segment.

```
1:  $i \leftarrow 10$ 
2:  $p \leftarrow 1$ 
3: while  $i \leq n + 5$  do
4:   if  $i$  is odd then
5:      $p \leftarrow p * i$ 
6:    $i \leftarrow i + 2$ 
```

- (1) Which one is the basic operation?

Solution: The comparison at line 4 or the addition at line 6.

- (2) Find the efficiency of the algorithm.

Solution: Let $t(n)$ be the number of comparisons. Then,

$$t(n) = \sum_{i=10,12,\dots,2k+10} 1 = \sum_{i=0}^k 1 = k + 1 \quad (1)$$

The last iteration of the condition $i \leq n + 5$ is $2k + 10 \leq n + 5$.

Estimated Answer: Assume $2k + 10 = n + 5 \implies k = \frac{n - 5}{2} \implies \therefore t(n) = \frac{n - 5}{2} + 1$.

Exact Answer:

$$2k + 10 \leq n + 5 < 2(k + 1) + 10$$

$$2k \leq n - 5 < 2(k + 1)$$

$$k \leq \frac{n - 5}{2} < k + 1$$

$$\therefore \lfloor \frac{n - 5}{2} \rfloor = k$$

Substitute in (1), $t(n) = k + 1 = \lfloor \frac{n - 5}{2} \rfloor + 1$.

2. Consider the following algorithm.

```
1:  $A \leftarrow 5$ 
2: for  $i \leftarrow 1$  to  $n$  do
3:   for  $j \leftarrow 1$  to  $i$  do
4:     for  $k \leftarrow i$  to  $n$  do
5:        $A \leftarrow A + 2$ 
```

- (a) Find the efficiency of the algorithm.

Solution: The basic operation is the addition at line 5. Quick analysis gives

$$\begin{aligned} t(n) &= \sum_{i=1}^n \sum_{j=1}^i \sum_{k=i}^n 1 = \sum_{i=1}^n \sum_{j=1}^i (n - i + 1) = \sum_{i=1}^n i(n - i + 1) \\ &= (n + 1) \sum_{i=1}^n i - \sum_{i=1}^n i^2 \\ &= (n + 1) \cdot \frac{n(n + 1)}{2} - \frac{n(n + 1)(2n + 1)}{6} \\ &= \frac{n(n + 1)(n + 2)}{6} \quad \leftarrow \text{after simplification} \end{aligned}$$

$$\therefore t(n) \in \Theta(n^3).$$

(b) Write the value of A as a function of n .

Solution: What are the values of A for each iteration:

Iteration	0	1	2	3	...	k
A	5	7	9	11	...	$5 + 2k$

Since the statement $A \leftarrow A + 2$ is repeated $t(n)$ times, then the value of A is equal to

$$A(n) = 5 + 2 \cdot t(n) = 5 + \frac{n(n+1)(n+2)}{3}$$

3. What is the time complexity of the following program segment?

```
count ← 0
for i ← 1 to n do
    j ← 1
    while j ≤ i do
        count = count + 1
        j ← 2j
```

4. Consider the following algorithm:

```
1: Algorithm Q1( $n$ )
2:   //  $n$  is any positive integer
3:    $a \leftarrow 1, b \leftarrow 2$ 
4:   for  $i \leftarrow 1$  to  $n$  do
5:      $j \leftarrow 2n$ 
6:     while  $j > 0$  do
7:       if  $j \leq n$  then
8:          $a \leftarrow 3a + b/8$ 
9:          $j \leftarrow j/3$ 
10:      else
11:         $b \leftarrow 5b + a/4$ 
12:         $j \leftarrow j - 1$ 
13:   return  $a + b$ 
```

- How many times the multiplication is executed in the statement of line 5.
- How many times the multiplication and addition are executed in the statement of line 7.
- Which one is the Basic Operation of the algorithm.
- Find the Efficiency of the algorithm and establish its order of growth.

5. Consider the following algorithm:

```
1:  $A \leftarrow 1$ 
2:  $B \leftarrow 2$ 
3: for  $i \leftarrow 1$  to  $n$  do
4:    $B \leftarrow 3B$ 
5:   for  $j \leftarrow 1$  to  $n$  do
6:     for  $k \leftarrow 1$  to  $j$  do
7:        $A \leftarrow A + 4$ 
```

- Find the number of times the assignment is executed in the statement ((b)).
- Find the number of times the testing is executed in the statement ((b)).
- Find the efficiency of the algorithm.
- Find B as a function of n .

6. Find the total number of multiplications and additions (together) as a function of n in the following algorithm.

```

1:  $s \leftarrow 0$ 
2: for  $j \leftarrow 4$  to  $2n$  do
3:   for  $k \leftarrow j^2$  down to 3 do
4:      $s \leftarrow s + k$ 
5:  $i \leftarrow 0$ 
6: while  $i \leq 4n$  do
7:    $q \leftarrow q + i^2$ 
8:    $i \leftarrow i + 2$ 

```

7. Consider the following algorithm

```

Algorithm Pow( $x, n$ )
  Input: a real number  $x$  and a positive integer  $n$ 
  if ( $n = 1$ ) then
    return  $x$ 
  else
    return  $x * \text{Pow}(x, n - 1)$ 

```

- (a) Trace the algorithm for $x = 5$ and $n = 4$.
 (b) What does the algorithm do?
 (c) Find the time complexity $T(n)$ and find its efficiency.
8. Consider the following algorithm

```

1: Algorithm Q( $n$ )
2:   // Input: A positive integer n
3:   if  $n = 1$  then
4:     return 3
5:   else
6:     return  $3 * Q(n/2) + 5n$ 

```

- (a) Write the recurrence relation for the algorithm.
 (b) Which one is the basic operation.
 (c) Write the recurrence relation for the time complexity $T(n)$.

Solution: The basic operation is multiplication at line 6.

- DIR(1) = 0, no multiplication.
- PRE-WORK(n) = 0 no multiplication.
- POST-WORK(n) = 1 (one multiplication).

$$\therefore T(n) = \begin{cases} 0 & \text{for } n = 1 \\ 0 + T(n-1) + 1 & \text{for } n > 1 \end{cases}$$

$$\therefore T(1) = 0, \quad T(n) = T(n-1) + 1.$$

- (d) Find the solution to $T(n)$.

Solution:

$$\begin{aligned}
 T(n) &= T(n-1) + 1 \\
 &= T(n-2) + 1 + 1 \\
 &= T(n-3) + 1 + 1 + 1 \\
 &\vdots \\
 &= T(n-k) + k
 \end{aligned}$$

Until $T(n - k) = T(1) \implies n - k = 1 \implies k = n - 1$.

$$\begin{aligned}\therefore T(n) &= T(1) + n - 1 \\ &= 0 + n - 1 \\ &= n - 1\end{aligned}$$

- (e) Establish the order of growth of the efficiency.

Solution: $T(n) \in \Theta(n)$.

9. Consider the following algorithm

```

1: Algorithm P( $n$ )
2:   if ( $n > 0$ ) then
3:      $p \leftarrow 1$ 
4:     for  $i \leftarrow 0$  to  $n - 1$  do
5:        $p \leftarrow p * 3$ 
6:       P( $n - 1$ )
7:       while  $n > 1$  do
8:          $n \leftarrow n/2$ 

```

- (a) Use *Detailed Analysis* to set up a recursive relation for the time complexity $T(n)$.
 (b) Use *Quick Analysis* to set up a recursive relation for the time complexity $T(n)$.
 (c) Find an explicit formula for $T(n)$ in part (b) and its efficiency.
 (d) Replace line 6 by P($n/2$) and repeat parts (a) to (c)

10. Integer multiplication can be defined as

$$MUL(m, n) = \begin{cases} m, & n = 1 \\ m + MUL(m, n - 1), & n > 1 \end{cases}$$

- (a) Write a recursive algorithm that multiplies integers using this equation.

Solution:

```

1: Algorithm MUL( $m, n$ )
2:   if ( $n = 1$ ) then
3:     return  $m$ 
4:   else
5:     return  $m + MUL(m, n - 1)$ 

```

- (b) Trace the algorithm for $m = 7$ and $n = 4$.

Solution:

$$\begin{aligned}MUL(7, 4) &= 7 + MUL(7, 3) \\ &= 7 + 7 + MUL(7, 2) \\ &= 7 + 7 + 7 + MUL(7, 1) \\ &= 7 + 7 + 7 + 7 \\ &= 28\end{aligned}$$

- (c) Give the time complexity recurrence relation for the number of additions and subtractions that are done by your algorithm.

Solution: The number of additions and subtractions is:

$$A(n) = \begin{cases} 0, & n = 1; \\ 1 + A(n - 1) + 1, & n > 1. \end{cases}$$

- (d) Solve the recurrence relation obtained in part (c) and find its efficiency class.